

A Square-Frequency Counterexample to a Laplace–Carleson Box Criterion

Run `fa_banach_001`, lane 0 (GPT5.6)

July 19, 2026

Source Problem

The source is Eskil Rydhe, *On Laplace–Carleson embeddings, and L^p -mapping properties of the Fourier transform*, arXiv:1907.11583 [1]. For an interval $I \subset \mathbb{R}$, write

$$Q_I = \{x + iy : x \in I, 0 < y \leq |I|\} \subset \mathbb{C}_+,$$

and define

$$\mathcal{L}f(z) = \int_0^\infty f(t)e^{2\pi itz} dt.$$

Section 5 asks the following question (PDF page 17):

5 A non-result for case (III)

If $\mathcal{L} : L^{3/2}(\mathbb{R}_+) \rightarrow L^{3/2}(\mathbb{C}_+, d\mu)$ is bounded, then μ satisfies

$$\mu(Q_I) \lesssim |I|^{1/2} \quad \text{for all intervals } I \subset \mathbb{R}. \quad (5)$$

Whether or not the converse holds is an open question, unless μ is sectorial, in which case the answer is positive.

In words: does the condition $\mu(Q_I) \lesssim |I|^{1/2}$ imply boundedness of

$$\mathcal{L} : L^{3/2}(\mathbb{R}_+) \longrightarrow L^{3/2}(\mathbb{C}_+, \mu)$$

for an arbitrary positive Borel measure μ ? The source proves sufficiency for sectorial measures and leaves the general case open.

Proof Intuition

The source itself introduces atoms at $n^2 + i$. Their real parts are sparse enough to obey the requested square-root box count, but their common imaginary part turns Laplace sampling into ordinary Fourier sampling after multiplying a test function by $e^{2\pi it}$. A finite trigonometric polynomial can then interpolate the first N atoms exactly. Its $L^{3/2}$ norm is at most its L^2 norm, namely $\sqrt{N}$, while the N unit samples have $\ell^{3/2}$ norm $N^{2/3}$. The mismatch $N^{1/6}$ proves unboundedness.

Result

Theorem 1. *Let $1 < p \leq q < 2$, let $p' = p/(p-1)$, and put $\alpha = q/p'$. There is a positive locally finite atomic measure μ on $\mathbb{C}_+$ such that*

$$\mu(Q_I) \lesssim |I|^{q/p'} \quad (I \subset \mathbb{R}),$$

but

$$\mathcal{L} : L^p(\mathbb{R}_+) \longrightarrow L^q(\mathbb{C}_+, \mu)$$

is not bounded.

Proof. Since $p \leq q < 2 < p'$, we have $0 < \alpha < 1$. Set

$$\beta = \frac{1}{\alpha} = \frac{p'}{q} > 1, \quad m_n = \lceil n^\beta \rceil, \quad \mu = \sum_{n=1}^{\infty} \delta_{m_n+i}.$$

The integers m_n are strictly increasing. Indeed, by the mean value theorem, $(n+1)^\beta - n^\beta > 1$ for every $n \geq 1$.

We first verify the box condition. If $|I| < 1$, then Q_I does not reach the horizontal line $\text{Im } z = 1$, so $\mu(Q_I) = 0$. Suppose that $L := |I| \geq 1$ and that I contains r members of (m_n) . If the first and last have indices a and b , then $b \geq a + r - 1$ and

$$\begin{aligned} L &\geq m_b - m_a \\ &\geq b^\beta - a^\beta - 1 \\ &\geq (a + r - 1)^\beta - a^\beta - 1 \\ &\geq (r - 1)^\beta - 1. \end{aligned}$$

Here the last inequality uses $(u+v)^\beta \geq u^\beta + v^\beta$ for $u, v \geq 0$ and $\beta \geq 1$. Therefore

$$r \leq 1 + (L+1)^{1/\beta} = 1 + (L+1)^\alpha \leq 3L^\alpha.$$

Thus $\mu(Q_I) \leq 3|I|^{q/p'}$ for every interval I .

For $N \geq 1$, define

$$f_N(t) = \mathbf{1}_{(0,1)}(t) e^{2\pi t} \sum_{k=1}^N e^{-2\pi i m_k t}.$$

These functions belong to $L^1(\mathbb{R}_+) \cap L^p(\mathbb{R}_+)$. At an atom of μ , the damping from the imaginary part cancels the factor $e^{2\pi t}$, and hence

$$\begin{aligned} \mathcal{L}f_N(m_n + i) &= \sum_{k=1}^N \int_0^1 e^{-2\pi i m_k t} e^{2\pi i(m_n+i)t} e^{2\pi t} dt \\ &= \sum_{k=1}^N \int_0^1 e^{2\pi i(m_n - m_k)t} dt \\ &= \begin{cases} 1, & 1 \leq n \leq N, \\ 0, & n > N. \end{cases} \end{aligned}$$

The last equality is exact because the m_n are distinct integers. It follows that

$$\|\mathcal{L}f_N\|_{L^q(\mu)} = N^{1/q}.$$

On the probability space $(0, 1)$, monotonicity of L^r norms and orthogonality give

$$\begin{aligned} \|f_N\|_{L^p(\mathbb{R}_+)} &\leq e^{2\pi} \left\| \sum_{k=1}^N e^{-2\pi i m_k t} \right\|_{L^p(0,1)} \\ &\leq e^{2\pi} \left\| \sum_{k=1}^N e^{-2\pi i m_k t} \right\|_{L^2(0,1)} = e^{2\pi} \sqrt{N}. \end{aligned}$$

Consequently,

$$\frac{\|\mathcal{L}f_N\|_{L^q(\mu)}}{\|f_N\|_{L^p}} \geq e^{-2\pi} N^{1/q-1/2} \longrightarrow \infty,$$

since $q < 2$. This proves the claimed unboundedness. $\square$

Corollary 1. *The open question in Section 5 of [1] has a negative answer. In the case $p = q = 3/2$, one may take*

$$\mu = \sum_{n=1}^{\infty} \delta_{n^2+i}.$$

This measure satisfies $\mu(Q_I) \leq 2|I|^{1/2}$, but $\mathcal{L} : L^{3/2}(\mathbb{R}_+) \rightarrow L^{3/2}(\mathbb{C}_+, \mu)$ is unbounded.

Proof. Here $p' = 3$, $\alpha = q/p' = 1/2$, $\beta = 2$, and $m_n = n^2$. The general proof applies. More explicitly, the test functions become

$$f_N(t) = \mathbf{1}_{(0,1)}(t) e^{2\pi t} \sum_{k=1}^N e^{-2\pi i k^2 t},$$

with

$$\|\mathcal{L}f_N\|_{L^{3/2}(\mu)} = N^{2/3}, \quad \|f_N\|_{3/2} \leq e^{2\pi} \sqrt{N}.$$

The ratio is at least $e^{-2\pi} N^{1/6} \rightarrow \infty$. $\square$

Verification Notes

The argument was checked step by step against the source normalization. At $z = n^2 + i$,

$$e^{2\pi i t z} = e^{2\pi i n^2 t} e^{-2\pi t},$$

so the damping cancellation has the required sign. Every test function is compactly supported and belongs to the operator domain. The estimate $\|S_N\|_p \leq \|S_N\|_2$ is in the correct direction because $(0, 1)$ has measure one and $p < 2$. Finally, translated intervals cause no counting issue: the increasing gaps of (m_n) make the maximal count occur near the beginning, and the displayed argument is uniform in the interval location.

The script `code/verify_square_counterexample.py` checks finite square counts and the exact finite sampling matrix. It is a sanity check only; the proof above is independent of computation.

Bounded Novelty Check

On July 19, 2026, the run indexes were searched for arXiv:1907.11583, the paper title, Laplace–Carleson embeddings, the $3/2$ endpoint, nonsectorial measures, square-frequency sampling, and the case-(III) box criterion. No duplicate or prior attempt was found. Web and arXiv searches used the same terms, the exact source sentence, $q < 2$, and the support $n^2 + i$.

The searches found the original paper, the earlier Laplace–Carleson theory of Jacob–Partington–Pott [2], and the later L^∞ /Orlicz work of Jacob–Partington–Pott–Rydhe–Schwenninger [3]. The latter cites the source and surveys adjacent regimes, but its new results concern essentially bounded or Orlicz inputs and output exponents $q \geq 2$; no statement of the counterexample above was found. This is a bounded search, not a guarantee of global novelty.

Limitations And Human Review

The theorem disproves sufficiency throughout $1 < p \leq q < 2$. It does not settle the remaining nonsectorial portion of the source’s case (III) with $q \geq 2$. Human review should focus on the box-count estimate and the exact Laplace/Fourier normalization; both are elementary and were independently checked.

References

- [1] E. Rydhe, *On Laplace–Carleson embeddings, and L^p -mapping properties of the Fourier transform*, Ark. Mat. 58 (2020), 437–457; arXiv:1907.11583.
- [2] B. Jacob, J. R. Partington, and S. Pott, *On Laplace–Carleson embedding theorems*, J. Funct. Anal. 264 (2013), 783–814; arXiv:1201.1021.
- [3] B. Jacob, J. R. Partington, S. Pott, E. Rydhe, and F. L. Schwenninger, *Laplace–Carleson embeddings and infinity-norm admissibility*, Anal. Math. Phys. 16 (2026), article 75; arXiv:2109.11465.